

What is removed at each stage

– magnitudes
(15 of 16 bits)

– 99% of
coordinates

– nothing:
measure survivors

– the runtime
hook

Full-precision
 ΔW (bf16)
never sized

One bit
per weight
sign + scale

1% of
coordinates
magnitude top-k

Effect
preserved
DEC C-ref

Merged
patch
no runtime hook

648 – 2560 MiB $\ddagger$
computed,
not measured

40.5 – 160.0 MiB
measured, $n = 6$
 ≤ 1.0007 bits/coord

0.41 – 1.60 MiB $\S$
payload only
+3.68 – 14.53 MiB
with support index

removal +0.353 $\dagger$
[+0.224, +0.493]
10/10 cells

llama-8B: 160.0 MiB dense $\rightarrow$ 1.60 MiB at 1% of coordinates — 100 $\times$ smaller